Turning Code into Compassion

A Visualize Aging Tool for Awareness and Reflection



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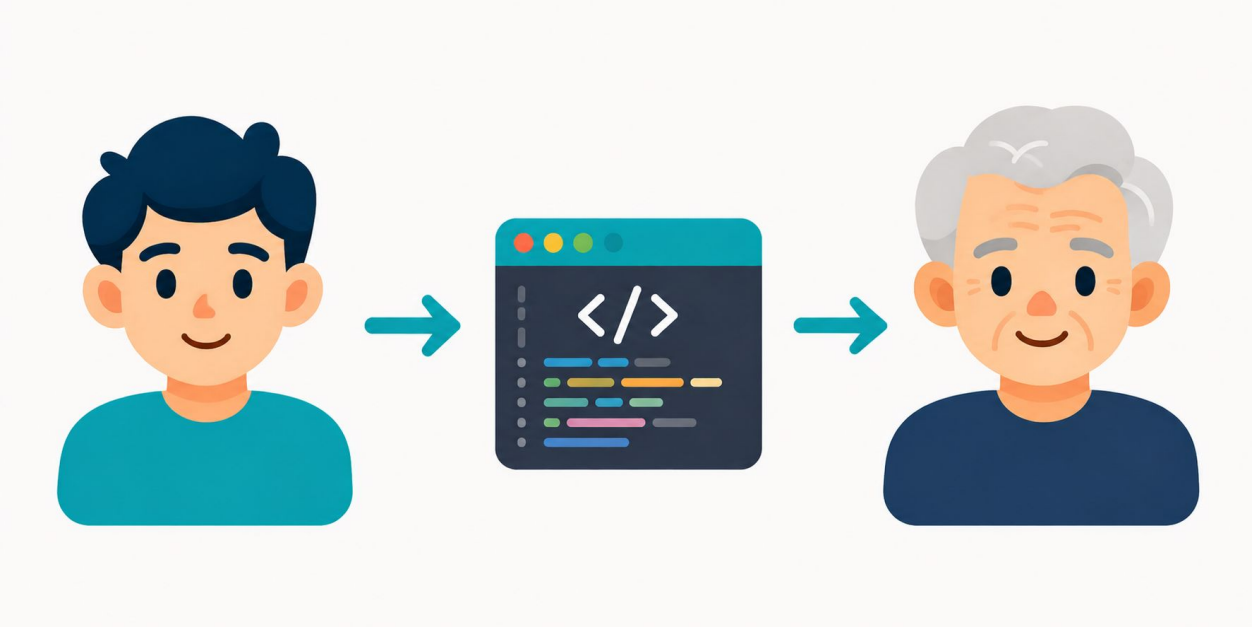
Dr. Deana McDonagh

(dis)ability DESIGN STUDIO

ILLINOIS
NCSA | National Center for Supercomputing Applications

Introduction

- Face Aging Tool
- Personalized Visualization
- Future Self Simulation



Significance



Encourage reflection and develop healthy living habits



Help young people experience and design with the perspectives of older adults



Promote healthy attitudes towards aging

Roadmap

Week 1

Orientation & Professional Development: Introduction to Modern Security Concerns

- Defined project goals and scope
- Conducted initial literature review
- Investigated existing face-aging tools
- Drafted preliminary design principles

Week 2

Professional Development: Research Presentations & Scientific Posters

- Expanded Literature Survey
- Face Aging Model Investigation
- Aging Pattern Research
- Framework & Presentation Development

Week 3

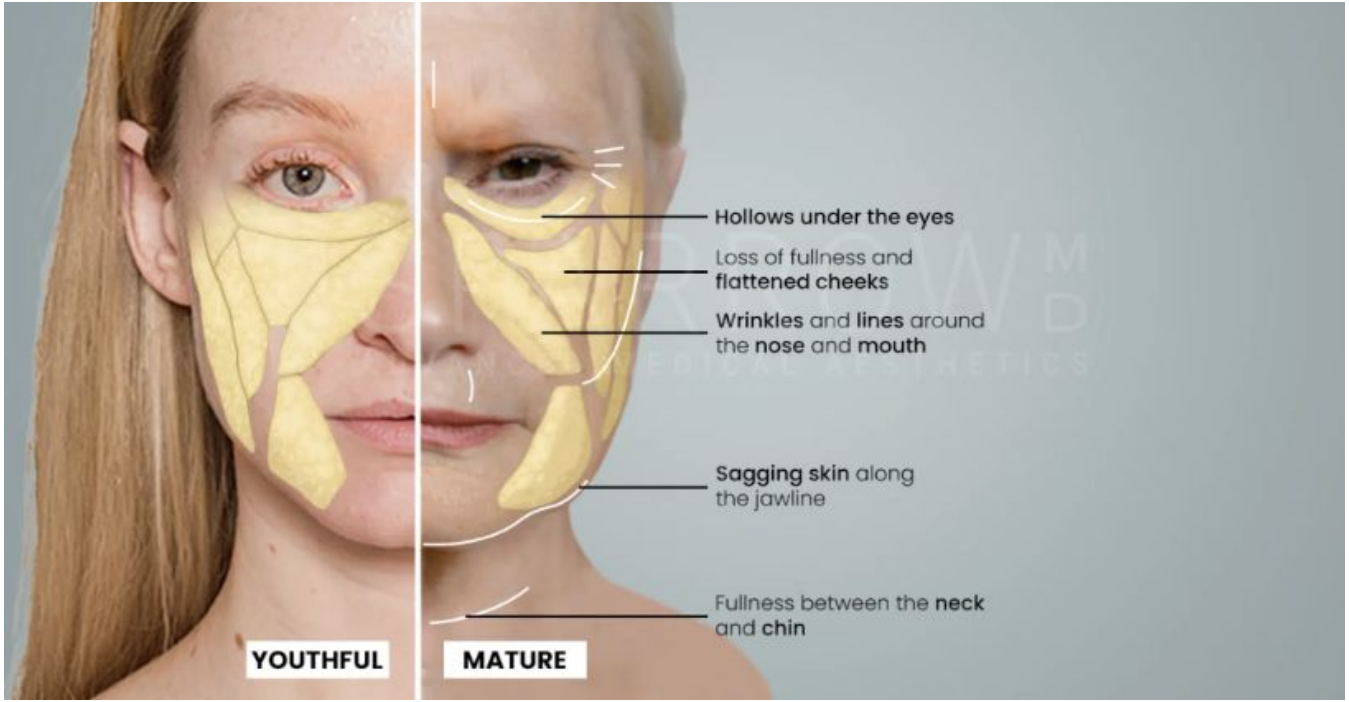
Professional Development: Scientific Writing

- Github Repository Setup
- Local Model Deployment
- Aging Knowledge Base Development
- Diffusion Pipeline Optimization



Intrinsic Aging

- Natural genetic process
- 5 most noticeable changes



Sparrow MD. "Facial Aging." *Sparrow MD*, sparrowmd.ca/facial-aging/. Accessed 30 June 2026.

Extrinsic Aging



Diet



Alcohol



Exercise



Smoking



Sleep



Pollution



Drugs



Finances



Sunlight



Stress



Dental

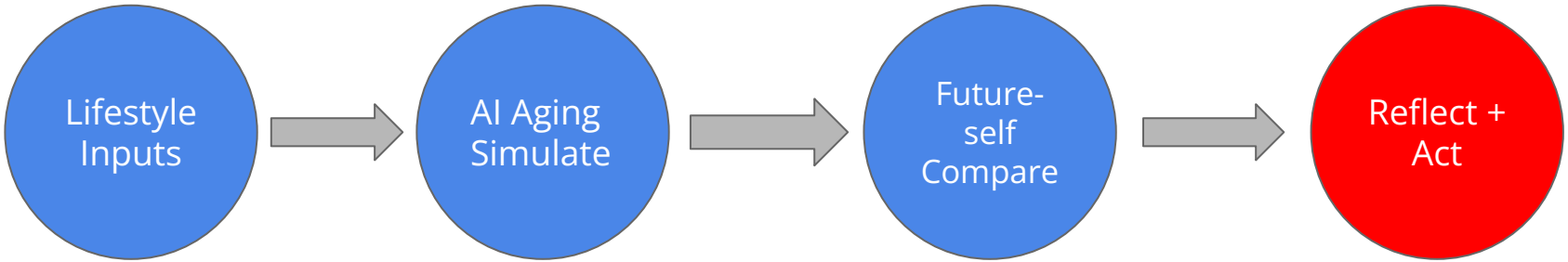


Skincare

Visualization Insights: Rationale

Aging Visualization becomes powerful when the user sees a plausible future version of themselves, not a generic warning.

Design Goal:
Make consequences visible, but still supportive and actionable.



Visualization Insights: Believability

- Aging must be recognizable.
- The visualization should look older, but still like the same person.
- Identity preservation is one of the most important requirements.
- A scenario selector is more honest than one fixed future image.

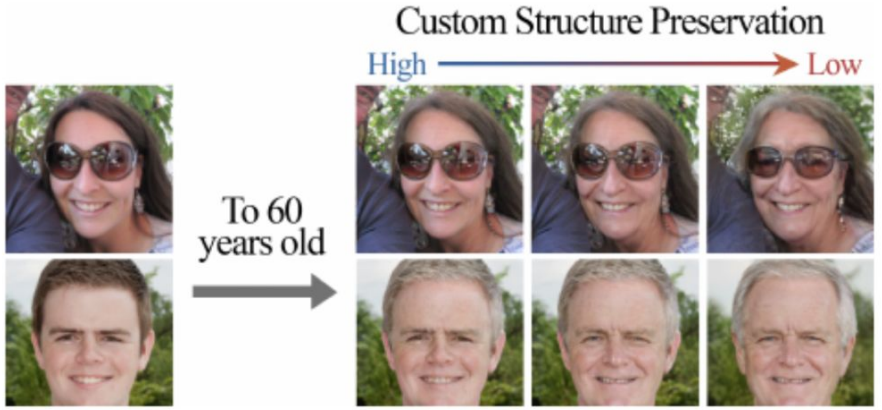


Figure Example: adjustable structure preservation changes how much faces shape is modified during aging.

Visualization Insights: Design Direction



- Make future consequences concrete
- Preserve Identity and diversity
- Compare scenarios, not certainties
- Support reflection and action

❌ “This is exactly what you will look like.”
✅ Different choices ➡️ different possible futures.

This is a scenario, not a prediction.

Software plan

- Engineering Optimization
- Model-level Optimization
- Software Intelligence



Preview and Outlook





References

Brinker, T. J., Seeger, W., Buslaff, F., et al. (2020). *A face-aging mobile app for skin cancer protection in secondary schools: A randomized clinical trial*. **JAMA Dermatology, 156**(7), 737–745. https://doi.org/10.1001/jamadermatol.2020.0360

Cajamarca, G., Herskovic, V., & Rossel, P. O. (2020). *Self-report and visualization of healthy aging for older adults over the internet*. **Sensors, 20**(15), 4348. https://doi.org/10.3390/s20154348

Chen, X., & Lathuilière, S. (2023). *Face Aging via Diffusion-based Editing (FADING).* arXiv:2309.11321.

Darejeh, A., & Singh, D. (2013). *A review on user interface design principles to increase software usability for users with less computer literacy*. **Journal of Computer Science, 9**(11), 1443–1450.

Flament, F., Abric, A., & Amar, D. (2015). *Gender-related differences in facial aging of Chinese subjects*. **Skin Research and Technology, 21**(3), 327–334.

Kim, B., et al. (2025). *Locally Controlled Face Aging with Latent Diffusion Models*. In **Proceedings of the ICCV 2025 Workshop on Perception Beyond the Visible Spectrum (P13N)**.

Lavie, T., & Tractinsky, N. (2004). *Assessing dimensions of perceived visual aesthetics of web sites*. **International Journal of Human–Computer Studies, 60**(3), 269–298.

Mercer, L. E., & McDonagh, D. (2014). *The pedagogy of discomfort: Transformational experiential learning*. In **DS 78: Proceedings of the 16th International Conference on Engineering and Product Design Education (E&PDE 2014)**.

Nielsen, J. (1994). *Enhancing the explanatory power of usability heuristics*. In **Proceedings of the SIGCHI Conference on Human Factors in Computing Systems** (pp. 152–158).

Norman, D. A. (2013). *The Design of Everyday Things* (Revised and Expanded ed.). Basic Books.

Wahid, J., Zhan, F., Rao, P., & Theobalt, C. (2024). *DiffAge3D: Diffusion-based 3D-aware Face Aging*. arXiv:2408.15922.

He, S., Liao, W., Yang, M. Y., Song, Y.-Z., Rosenhahn, B., & Xiang, T. (2021). Disentangled Lifespan Face Synthesis. IEEE/CVF International Conference on Computer Vision (ICCV).

Burgess, I. R., Owen, A., Scholtens, K., & Grogan, S. (2025). Men's experiences of a personalised, appearance-based, facial-morphing, safer drinking intervention. Journal of Health Psychology, 30(2), 156–170.



Thank you

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